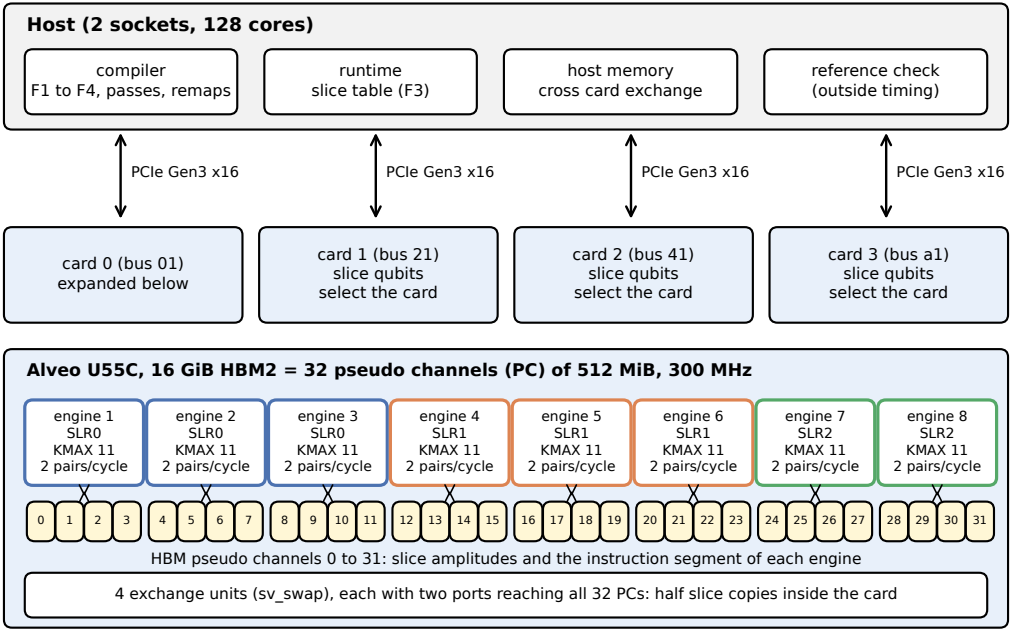
Host (2 sockets, 128 cores)
compiler
F1 to F4, passes, remaps
runtime
slice table (F3)
host memory
cross card exchange
reference check
(outside timing)
PCIe Gen3 x16
PCIe Gen3 x16
PCIe Gen3 x16
PCIe Gen3 x16
card 0 (bus 01)
expanded below
card 1 (bus 21)
slice qubits
select the card
card 2 (bus 41)
slice qubits
select the card
card 3 (bus a1)
slice qubits
select the card
Alveo U55C, 16 GiB HBM2 = 32 pseudo channels (PC) of 512 MiB, 300 MHz
engine 1
SLR0
KMAX 11
2 pairs/cycle
engine 2
SLR0
KMAX 11
2 pairs/cycle
engine 3
SLR0
KMAX 11
2 pairs/cycle
engine 4
SLR1
KMAX 11
2 pairs/cycle
engine 5
SLR1
KMAX 11
2 pairs/cycle
engine 6
SLR1
KMAX 11
2 pairs/cycle
engine 7
SLR2
KMAX 11
2 pairs/cycle
engine 8
SLR2
KMAX 11
2 pairs/cycle
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31
HBM pseudo channels 0 to 31: slice amplitudes and the instruction segment of each engine
4 exchange units (sv_swap), each with two ports reaching all 32 PCs: half slice copies inside the card